

An LLM-Guided Tutoring System for Social Skills Training

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Social Skills



Why Social Skills are More Critical Than Ever

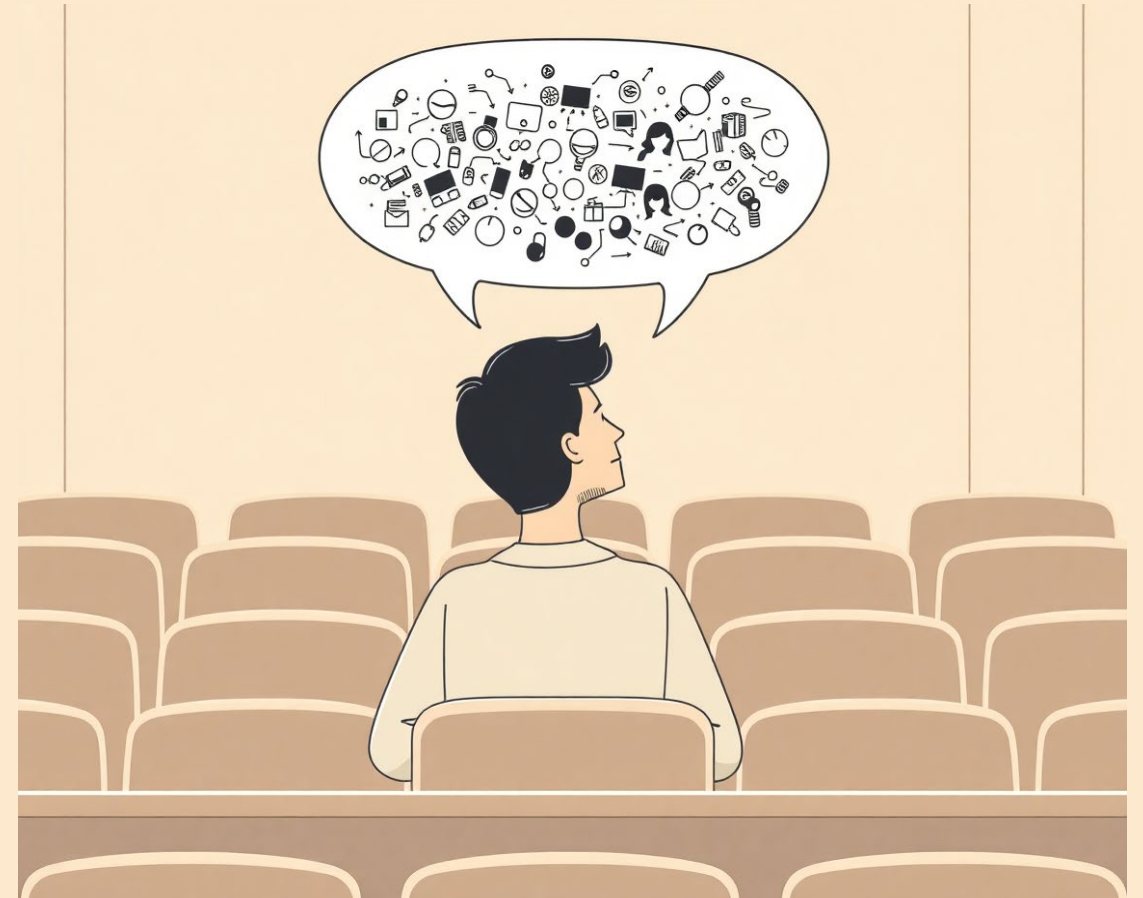
- In today's fast-paced, digital world, skills like communication, collaboration and empathy - often called '*soft skills*' or '*social skills*' are no longer just nice to have.
- They are essential for success, driving employee engagement and boosting productivity.
- Think of them as the operating system for effective teamwork.

The Limits of Traditional Training

The Classroom Problem

Traditional methods for teaching social skills, like classroom lectures or workshops, often fall short. They provide information but lack the messy, unpredictable nature of real-life conversations.

- One-size-fits-all curriculum.
- Lacks dynamic, real-world context.
- Limited opportunities for personal practice.
- Difficult to simulate emotionally charged situations.



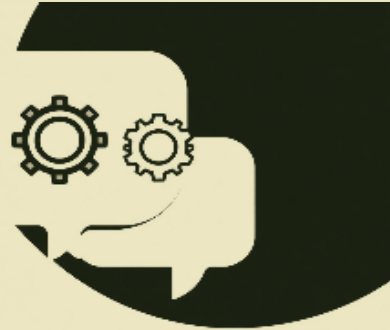
A New Era of Learning: AI as a Practice Partner

- Recent breakthroughs in LLMs the advanced AI that powers tools like ChatGPT, offer a revolutionary solution.
- These models can understand, generate, and adapt conversation, creating realistic, interactive training environments.



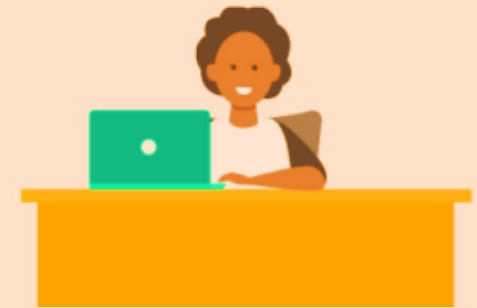
Dynamic Narratives

LLMs can create branching, evolving storylines for role-playing, making conversations feel alive and responsive



Realistic Simulations

AI can simulate complex human behaviors and social interactions in real-time, creating a believable practice field.



Personalized Agents

By giving AI agents unique personas, training becomes more engaging and realistic, mirroring diverse personalities learners will encounter.

Foundations: Understanding the Key Concepts

Social Skills Training (SST)

This is a structured approach to improving the key behaviors needed for successful social interactions. It's like a coaching program for communication.

- Instruction (Learning the 'how-to')
- Modeling (Watching examples)
- Behavior Rehearsal (Practice)
- Feedback (Getting pointers)
- Reinforcement (Building confidence)

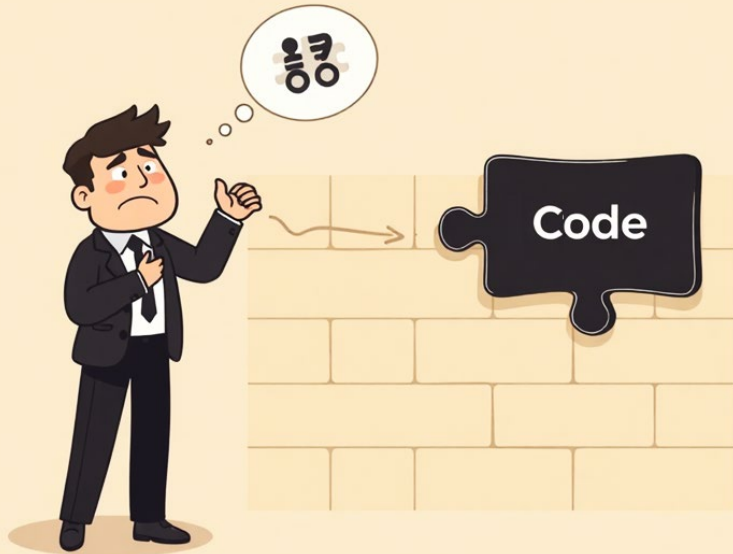
Intelligent Tutoring Systems (ITS)

Think of an ITS as a personal digital tutor. It's a computer system that provides customized instruction or feedback to learners, adapting to their pace and needs.

- Used to teach counseling skills.
- Assists people with social challenges.
- Develops interpersonal skills.

The Problem We're Addressing

Technical skills



The Challenge with Existing AI Tutors

- While Intelligent Tutoring Systems for social skills exist, they have a major bottleneck: creating the content.
- Building and modifying training scenarios has traditionally required programming expertise and significant effort
- Making them rigid and hard to personalize.

Solution: Introducing GLOSS

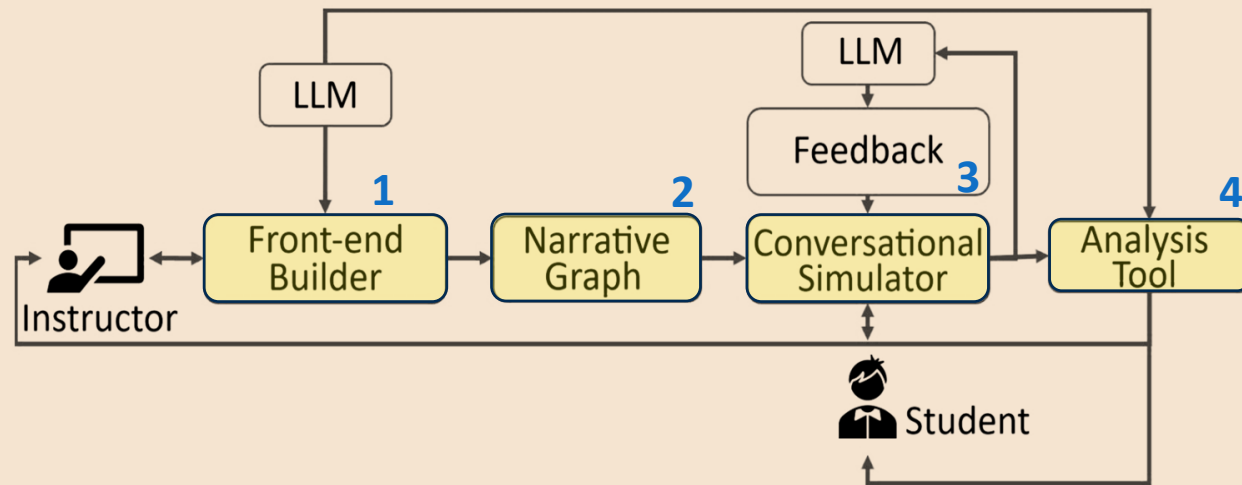


Guided Learning for Optimizing Social Skills

We introduce the Guided Learning for Optimizing Soft Skills (GLOSS) framework.

- It's a system that empowers instructors to collaborate with LLMs
- Making it easy to create powerful, dynamic, and personalized social skills training.
- Instructor-in-the-loop approach to rapidly build and modify domain-specific exercises
- On-demand training with immediate feedback and evaluation
- User performance analysis and recommendation for improvement

The GLOSS Framework: How It Works



GLOSS is composed of **4** interconnected components that create a seamless loop from content creation to student practice and analysis.

Component 1: The Front-end Builder

Putting the Instructor in the Director's Chair

The builder is a tool that lets instructors design training scenarios without writing a single line of code. It's a true collaboration between human expertise and AI creativity.



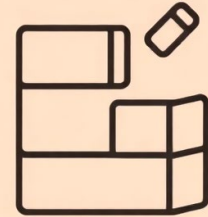
Use a Template

Start with a pre-built scenario with scripted dialogue, like a common customer service issue.



Generate with AI

Simply write a prompt and let the LLM generate a starting point.

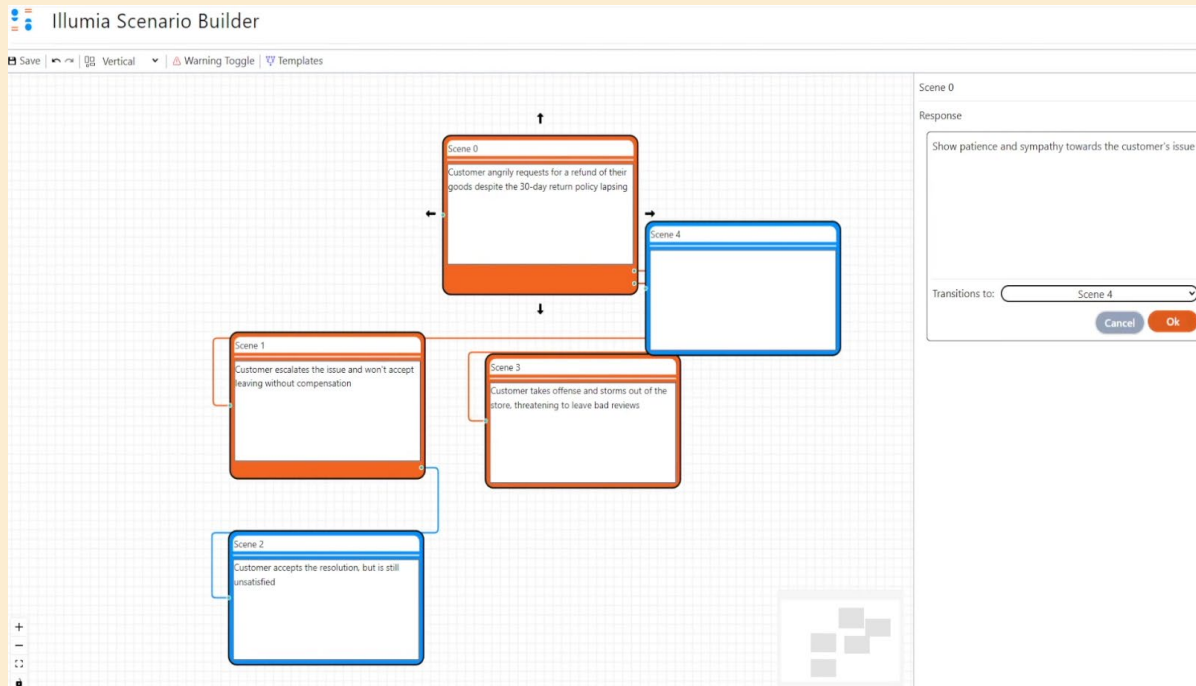


Create from Scratch

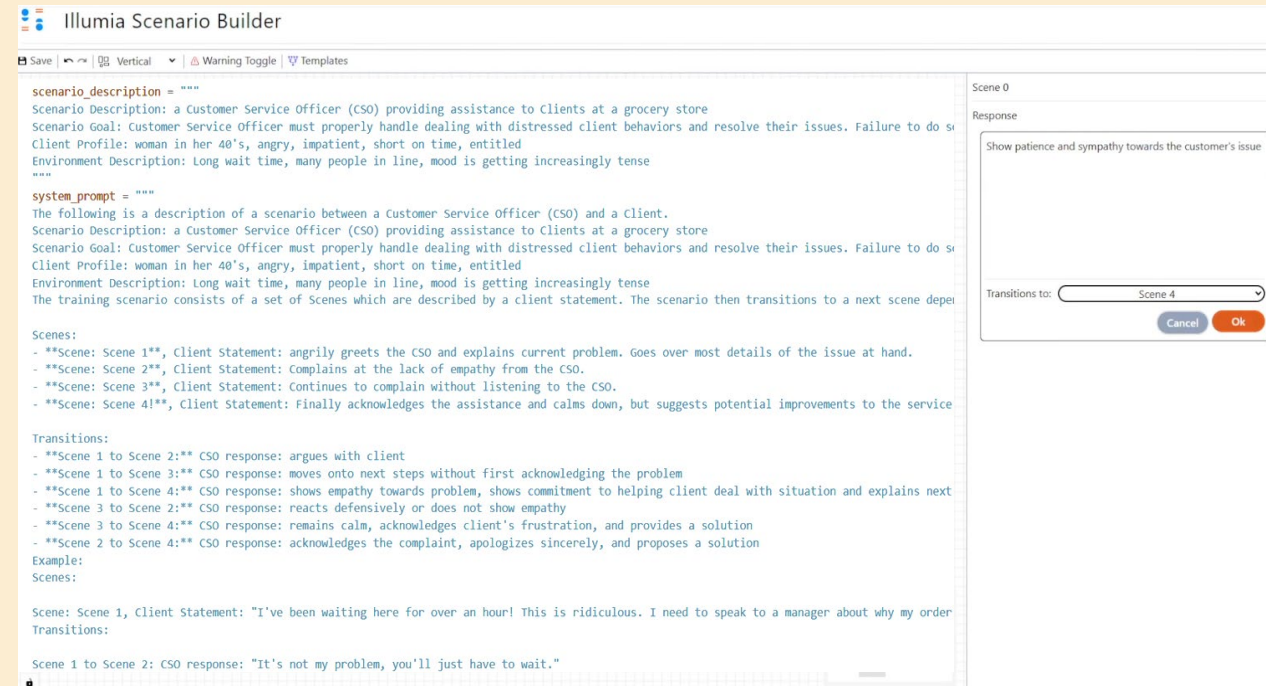
Manually build a scenario node by node for complete control over the interaction.

Component 1: The Front-end Builder

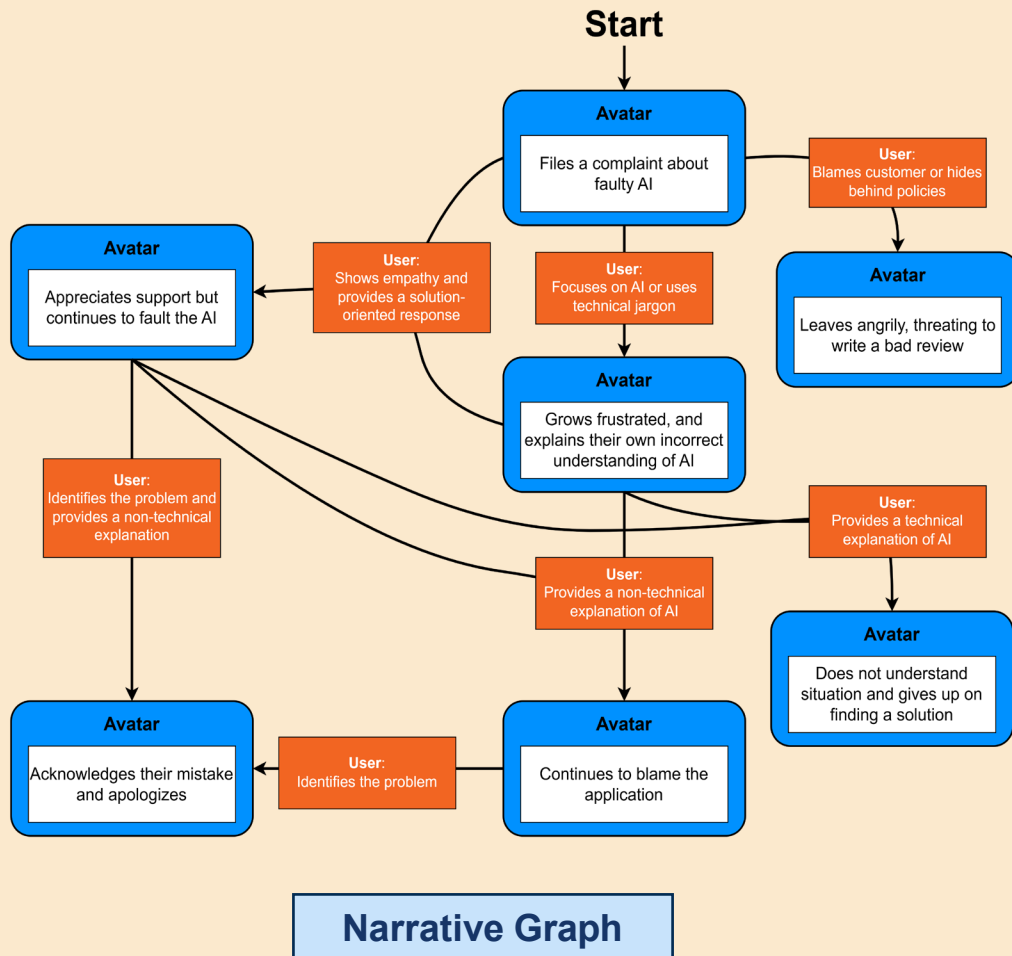
Scenario builder with templates or manually from scratch



Scenario builder with AI Generation



Component 2: The Narrative Graph



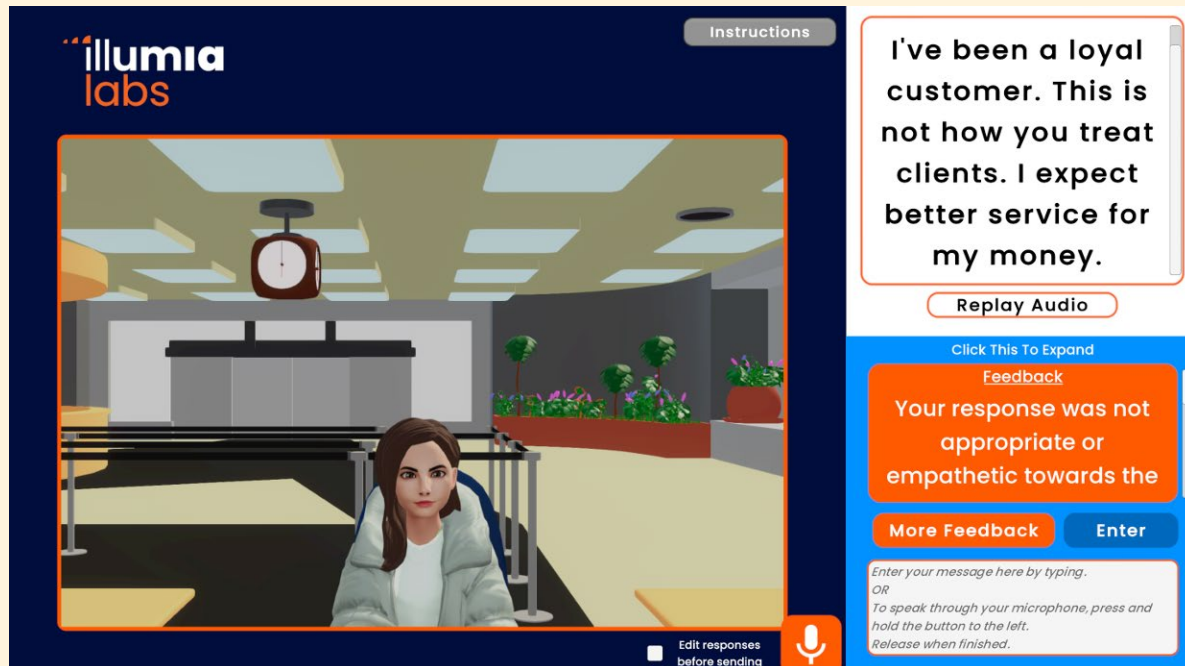
Visualizing the Conversation's 'What Ifs'

It's a visual representation of all the possible paths an interaction can take.

- **Nodes (Scenes):** Each circle represents a moment or a turn in the conversation (e.g., the customer's initial complaint).
- **Edges (Transitions):** The lines connecting the nodes represent the student's choices and the resulting reactions (e.g., responding politely leads to a calmer customer).

- Instructors can easily add, edit, or delete parts of the graph to tailor the complexity to the learner's skill level.

Component 3: The Conversational Simulator



Conversational Simulator

The Practice Field for Social Skills

- This is the interface where students actively practice. It's like a flight or racing simulator, but for difficult conversations.
- Students can type or speak their responses and interact with an AI-powered avatar that reacts based on the narrative graph.
- It provides a safe environment to rehearse uncomfortable or emotionally charged situations without real-world consequences.

The Key: Dynamic Path Generation

What if a user says something completely unexpected?

- GLOSS's most powerful feature is its ability to adapt in real time.

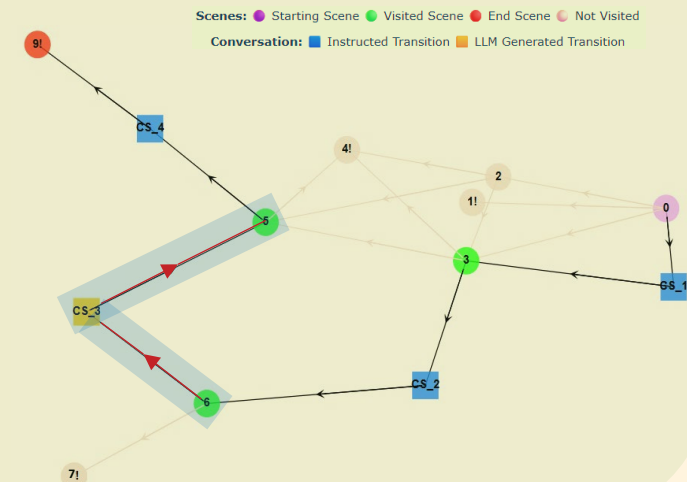
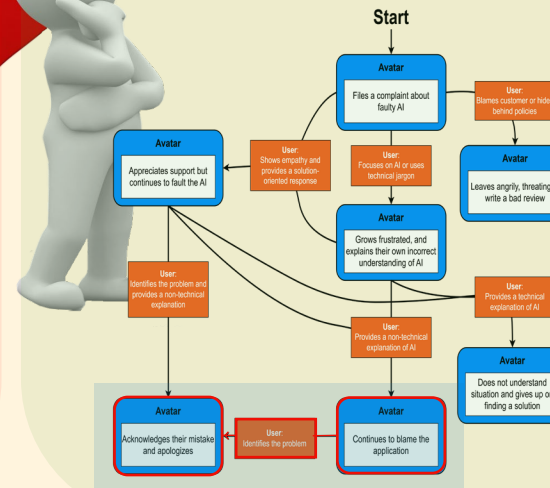
Traditional Systems

If a student's response doesn't match a pre-defined option, the system hits a dead end or gives a generic 'I don't understand' reply.

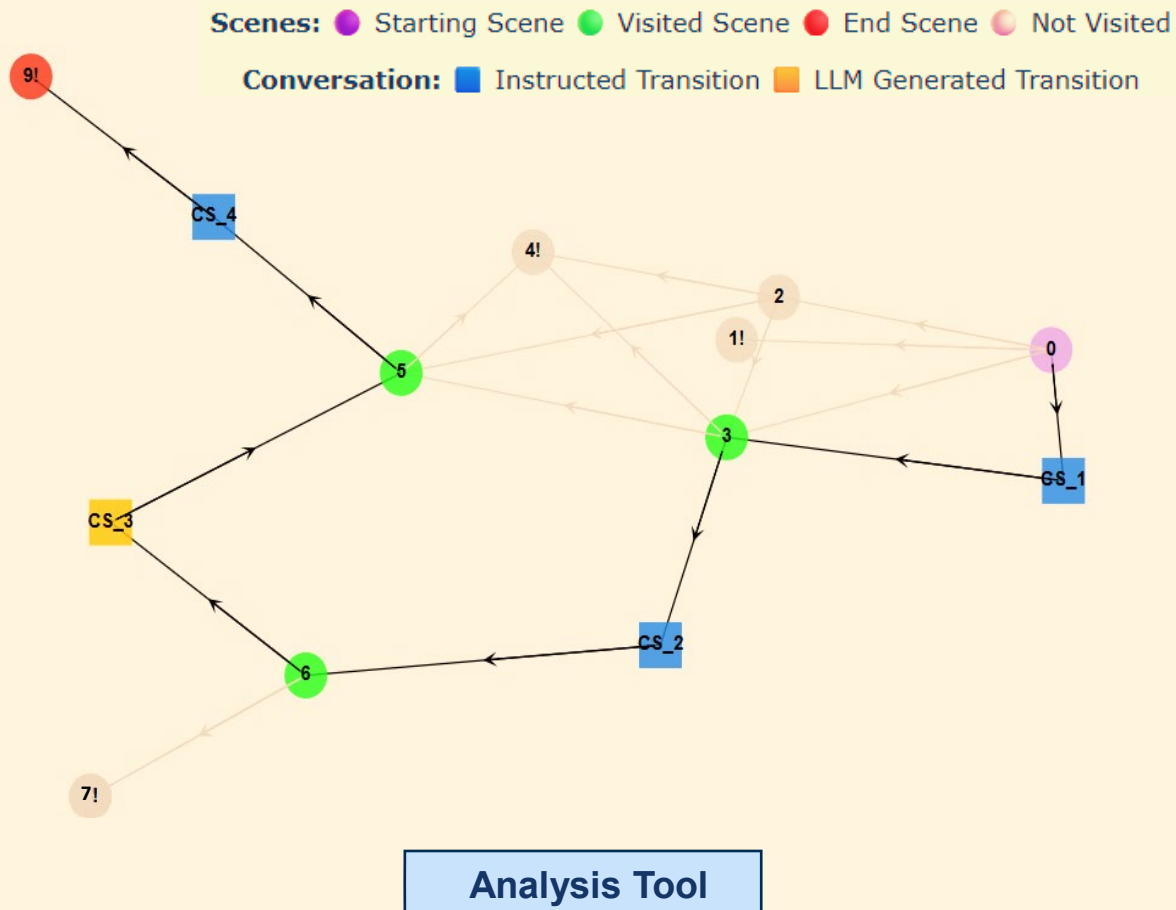


GLOSS

The LLM analyzes the student's intent and dynamically generates a new, logical branch in the narrative graph on the fly, allowing the conversation to continue seamlessly.



Component 4: Analysis Tool



This tool visualizes the student's conversation path through the narrative graph. It serves two crucial purposes:

For Instructors:

- Assess student performance.
- Check if AI-generated content meets goals.
- Identify areas to improve the scenario.

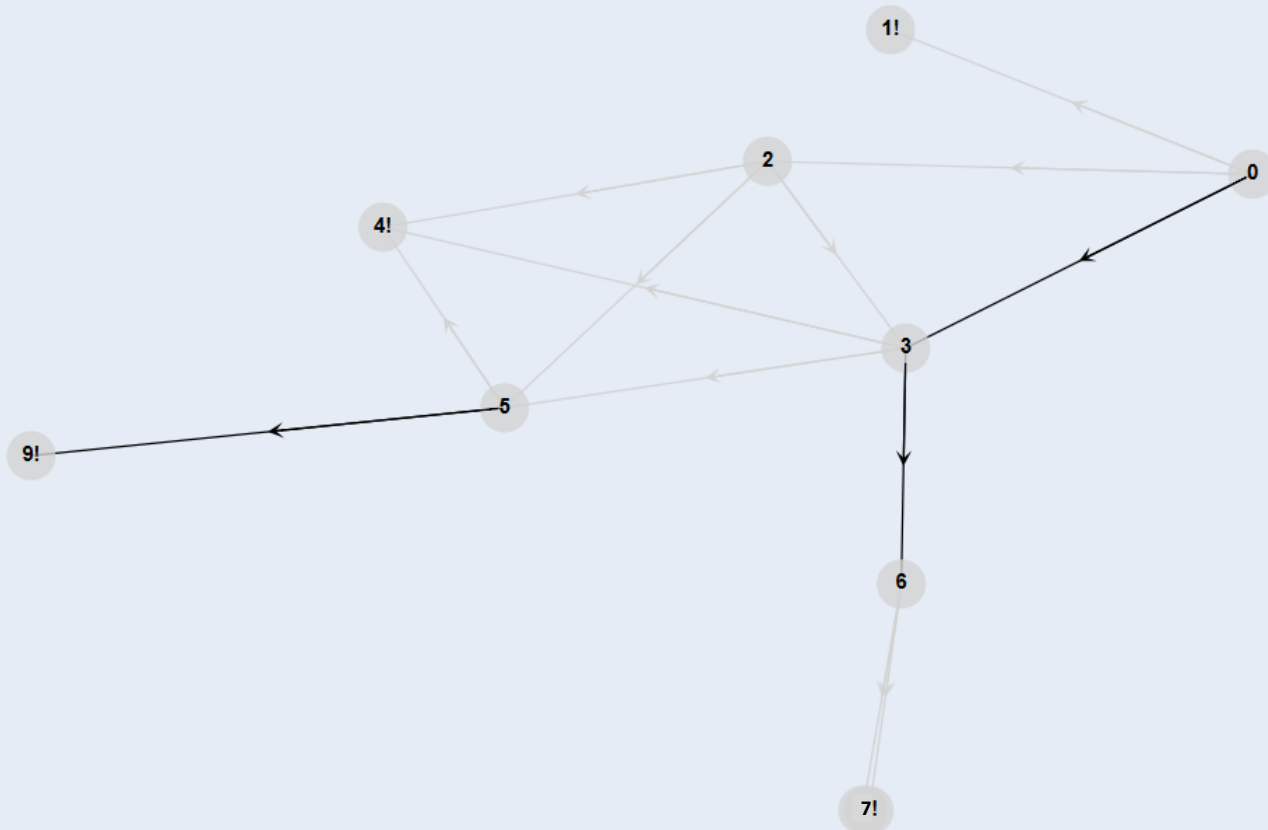
For Students:

- Reflect on their choices and outcomes.
- Identify patterns in their communication.
- Improve performance in future simulations.

Component 4: Analysis Tool



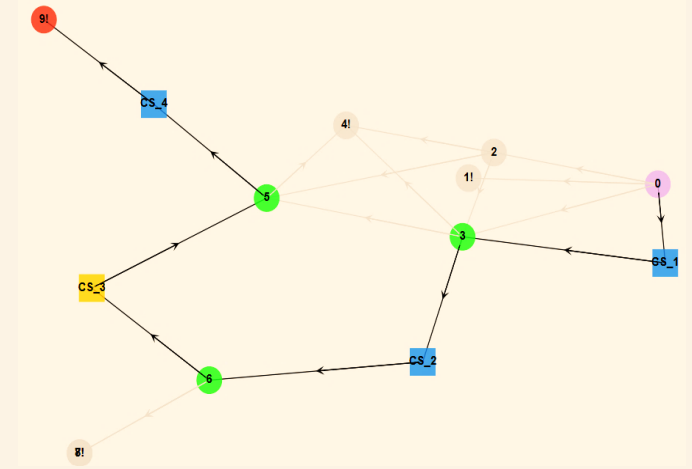
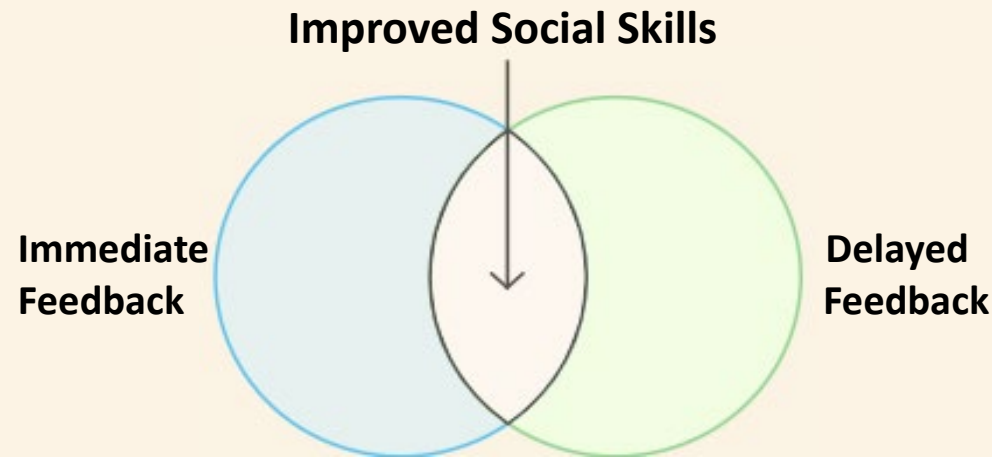
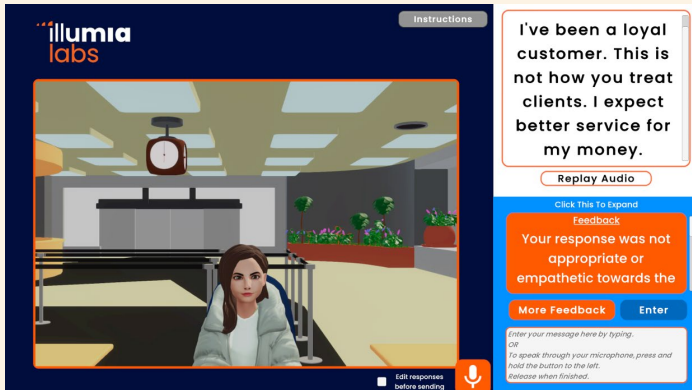
Scenario Transitions Graph



**Initial Graph
after
instructor's
scenario setup**

Learning Through Feedback

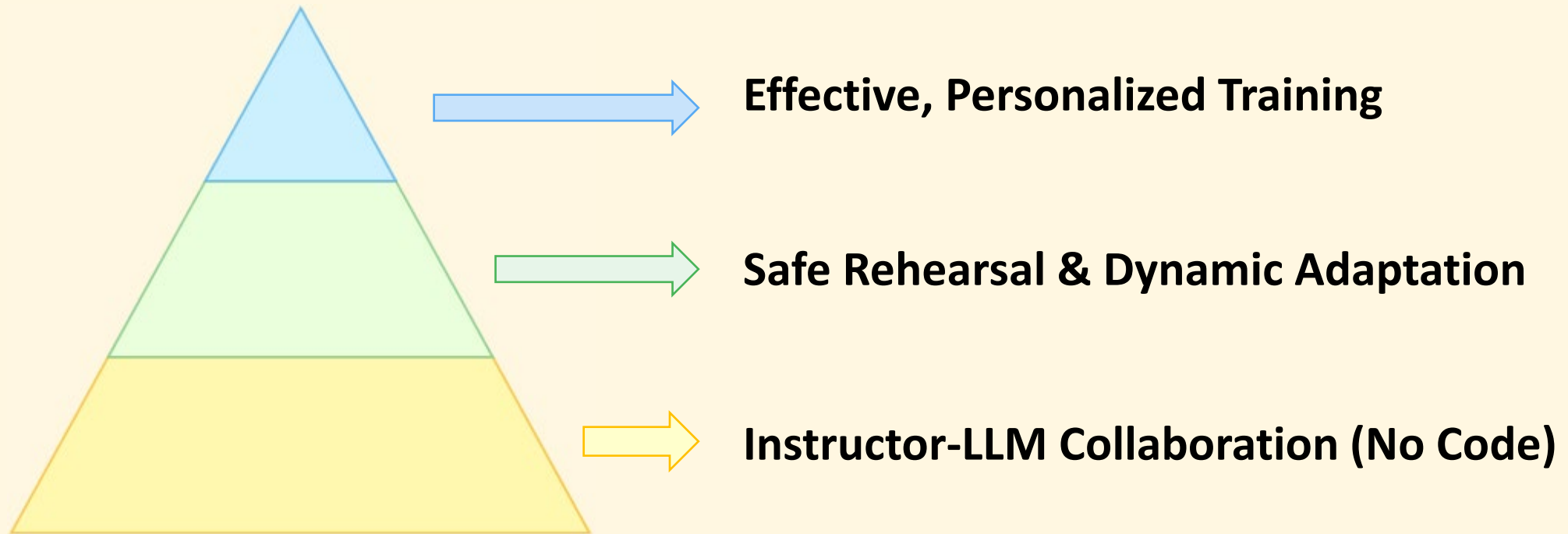
Effective learning requires feedback. GLOSS provides it in two powerful ways, catering to both in-the-moment adjustments and long-term reflection.



Immediate Feedback: After a student responds, an independent LLM prompt provides instant critique and suggestions, allowing them to get feedback immediately.

Delayed Feedback: The Analysis Tool provides a post-session review, helping students and instructors see the bigger picture.

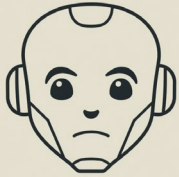
The GLOSS Advantage: Key Takeaways



GLOSS transforms social skills training by making it more accessible to create, more adaptive for learners, and ultimately, more effective at building real-world communication skills.

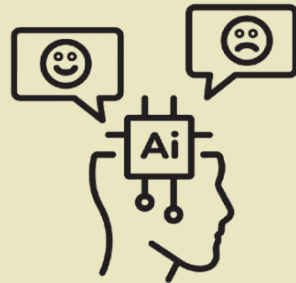
The Road Ahead: Future Work

We plan to continue refining the system to make it even more powerful and effective.



Avatar Responses Enhancement

Improving the AI avatar's responses and reactions to be even more lifelike and emotionally intelligent



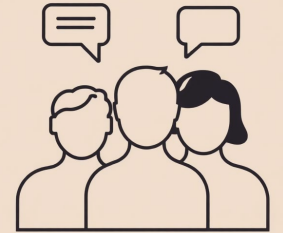
Emotion Detection

Enhancing the avatar's facial expressions using emotion detection from the conversation statements



Refine Feedback Mechanisms

Further developing the immediate and delayed feedback to maximize learning impact.



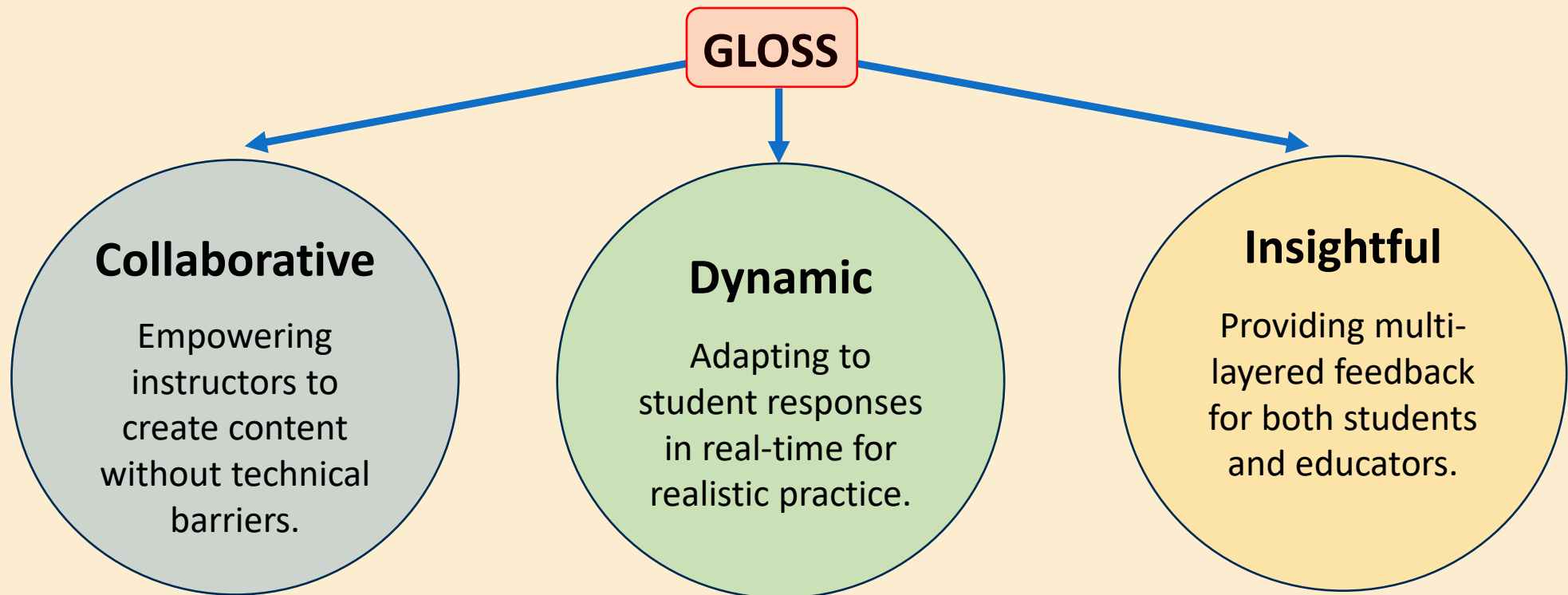
Conduct User Studies

Testing with real students and instructors to measure its effectiveness and gather insights for improvement.

Conclusion

The GLOSS framework represents a significant step forward in social skills training.

By creating a synergistic loop between instructors, learners, and Large Language Models, we can build training that is:



Acknowledgement



- This work has been completed for Illumia Labs Inc. as part of their advanced training platform.
- Part of this work has taken place in the Intelligent Robot Learning (IRL) and EdTeKLA Lab at the University of Alberta
- Collaborations from WASI Lab at New Mexico State University and University of Michigan-Dearborn.
- Supported in part by grants from Amii, a Canada CIFAR AI Chair, DRAC, Mitacs and NSERC.



Thank you!

For more information:

Paper: <https://doi.org/10.1609/aaai.v39i28.35353>

Video: <https://openreview.net/attachment?id=GMXZvXfSYh&name=video>